Lab 42: Try tool_choice three ways and read the failures

Topic 4: Prompt Engineering & Structured Output · Task 4.3: Enforce structured output using tool use and JSON schemas

Runs on Simulator 6: Structured Data Extraction · Self-practice

Source: Built from exam guide task statement 4.3 (no practice question covers this angle)

Commit to an answer first

Extraction has to come back as a tool call, and you don't know the document type in advance. How do you control whether Claude calls a tool?

A. Know the options, and test against your model.

B. Assume any forced setting works on every model.

Simulate the solution

```
# In the browser: open Simulator 6, then Lab 42
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 42

# In a terminal, from simulators/simulator-6-structured-extraction/
uv run extract.py --tool-choice any --max-docs 2
uv run extract.py --tool-choice forced --max-docs 2
```

What you should see

The Agent SDK has no tool_choice setting, so the script asks for it in the prompt. Read what Claude did, and the lesson in the evidence.

Open `simulators/simulator-6-structured-extraction/evidence/` and find it yourself.

The answer

A. Know the options, and test against your model.

Now change one thing

Change DOC-04's printed total in fixtures/invoices.txt, rerun, and watch by_document_type.invoice fall while the others hold.